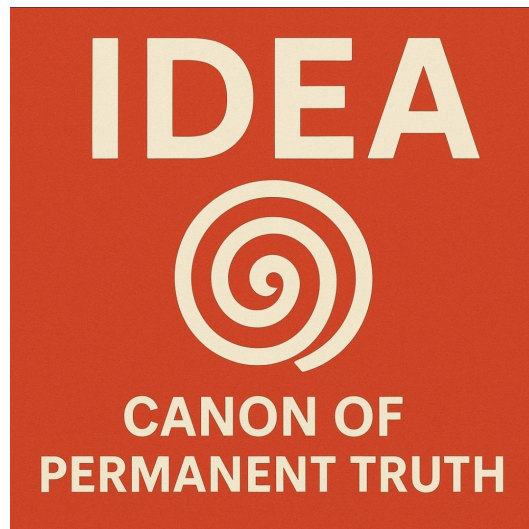


# Canon of Permanent Truth

## IDEA

Version 2.1: corrected mathematics and dependent claims

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*Stability is not the absence of change but the return through it.*

### INITIATE   DEFINE   ENGAGE   ALIGN

This revision preserves IDEA's four phases and four perspectives. It replaces the defective formal construction in v2 with a model of contrast, explicit identity criteria, admissible transformation-and-return protocols, and closure under composition.

The mathematical results concern the stated model. The broader claim that IDEA is the necessary form of all reality remains a philosophical hypothesis; it is not established by the equations.

## Revision note and abstract

The original five-page *Canon of Permanent Truth (IDEA)* provides the conceptual sequence and the perspectives Self, Relation, Object, and System. The 28-page v2 attempts to derive that architecture and universal truth from recursion. This corrected edition retains the architecture, repairs the formulas throughout, and revises the conclusions that depended on invalid mathematics.

The changes are substantive. In particular, v2's definition of recursive functions actually describes finite-order permutations; its phase hierarchy changes mathematical types; its unrestricted fixed-point condition is empty on every state space with at least two elements; and its final theorem uses non-return where it requires return. Several examples also confuse truth, consistency, recurrence, and evidence.

### Abstract

IDEA is presented here as a parameterized framework for testing whether a specified identity survives specified transformations and returns. INITIATE records distinctions; DEFINE states structural constraints and the identity to be tracked; ENGAGE evaluates declared tests; ALIGN establishes invariance under a declared family of return protocols and their finite compositions. The four perspectives organize these tests without being asserted to follow uniquely from recursion.

A common state space keeps all operations well typed. A feature map makes preservation of identity explicit. Return protocols are specified independently of their success, and universal claims are restricted to the declared protocol family. A closure theorem proves that invariance under generating return maps extends to every finite composite when those maps respect the chosen identity equivalence. Finite termination is established only under a finite-state assumption. The sixteen original examples are retained with corrected logical and evidential assessments.

### How to read this edition

- **Definitions** specify the framework. They do not prove that it is necessary or uniquely correct.
- **Assumptions** describe what an application must supply and what a theorem requires.
- **Propositions and theorems** include arguments establishing their stated conclusions.
- **Interpretive passages** preserve IDEA's conceptual intent without treating it as a demonstrated universal ontology.

In this edition, *permanence* means preservation of a declared identity under every finite composition of declared return protocols. It does not mean empirical certainty, indestructibility under arbitrary interventions, or truth in every possible formal system.

# 1. Mathematical setting and notation

Let  $U$  be a set with at least two elements, used to represent a common background of possible distinctions. Calling it a *primordial field* is an interpretation. A set already comes with membership and equality; set notation does not derive distinction from a literally structureless origin.

Let  $X$  be a nonempty set of candidate states or structured descriptions. It is separate from the set of transformations on those states. If a model contains persons, objects, relations, and systems, their encodings must be distinguished within  $X$ , for example by type tags and context records.

For a set  $S$ , write  $\mathcal{P}(S)$  for its power set. Let

$$\text{End}(X) = \{f \mid f : X \rightarrow X\}.$$

These are endofunctions, not necessarily computable, invertible, periodic, or successful at preserving identity. All compositions below use the same  $X$ .

| Symbol                                  | Meaning and type                                                                   |
|-----------------------------------------|------------------------------------------------------------------------------------|
| $U$                                     | Background set for contrast witnesses.                                             |
| $X$                                     | Candidate state space; $x \in X$ .                                                 |
| $\Delta$                                | Unequal ordered pairs in $U \times U$ .                                            |
| $Q$                                     | Four chosen perspective labels.                                                    |
| $\kappa : X \rightarrow K$              | Declared identity features; $x \equiv_{\kappa} y$ means $\kappa(x) = \kappa(y)$ .  |
| $\mathcal{P}_q$                         | Declared pairs $(f, r)$ of endofunctions: a transformation and a candidate return. |
| $\mathcal{E}_q \subseteq \mathcal{P}_q$ | Nonempty subset selected for engagement tests.                                     |
| $H_q$                                   | Completed protocol maps $r \circ f$ for perspective $q$ .                          |
| $\mathcal{M}$                           | Monoid generated by all completed protocol maps.                                   |
| $B \subseteq X$                         | States satisfying all initiation and definition requirements.                      |
| $I_* \subseteq X$                       | States in $B$ preserving identity under every map in $\mathcal{M}$ .               |
| $N_* \in \mathbb{N}_0$                  | First finite stabilization index, <i>when it exists</i> .                          |

The symbols  $I_*$  and  $N_*$  replace the two incompatible uses of  $I_{\text{final}}$  in v2. The phase filters  $\nabla, \mu, \varepsilon, \alpha$  all have type  $\mathcal{P}(X) \rightarrow \mathcal{P}(X)$ ; they are never simultaneously treated as state maps and sets.

The model uses ordinary classical mathematics. Applying IDEA to a different logic requires an explicit replacement of the relevant logical semantics.

## 2. Four perspectives as an explicit architecture

Retain the two descriptive axes interior/exterior and singular/plural. Their product supplies four labels:

$$Q = \{\text{interior, exterior}\} \times \{\text{singular, plural}\}.$$

Use  $q_0 = I$ ,  $q_1 = We$ ,  $q_2 = It$ , and  $q_3 = Its$  with the following correspondence.

| Label | Perspective        | Interpretive role                                                             |
|-------|--------------------|-------------------------------------------------------------------------------|
| $I$   | Interior, singular | Self: the represented standpoint or internal organization of one participant. |
| $We$  | Interior, plural   | Relation: shared interpretation, coordination, and mutual expectations.       |
| $It$  | Exterior, singular | Object: the observable behavior or structure of an individual entity.         |
| $Its$ | Exterior, plural   | System: interactions, constraints, and organization across a composite.       |

The count  $|Q| = 2 \cdot 2 = 4$  follows *once these two binary axes are chosen*. The choice and completeness of those axes do not follow from the return equations.

The four proposed return motifs in v2 are useful descriptions but are not disjoint mathematical classes. If  $r(a) = b$  and  $r(b) = a$ , that mutual return is also a two-step cycle. Taking  $f = r$  gives  $r(f(a)) = a$ , so it is simultaneously an instance of transformative return. A map can also have fixed points elsewhere. Fixed-point behavior alone therefore does not assign every map to exactly one of the four labels.

Here, each perspective receives an explicitly specified family of tests. A test may belong to more than one perspective. Thus  $H_I, H_{We}, H_{It}, H_{Its}$  may overlap; they are not asserted to partition all transformations. Labels are not state spaces, and expressions such as  $\text{Rec}(We)$  are not used.

### What a fourfold derivation would still require

A derivation would need independently stated primitives, an equivalence relation on candidate phases or scopes, a rule of generation, a proof of exhaustiveness, and a proof that no smaller collection suffices. None of these conclusions follows simply by naming four return patterns. This edition uses the fourfold organization as a framework assumption, preserving it as a subject for later investigation.

### 3. Return protocols and the identity being tested

Choose a feature map  $\kappa : X \rightarrow K$ . It states what is to count as the identity of a candidate during this assessment:

$$x \equiv_{\kappa} y \iff \kappa(x) = \kappa(y). \quad (1)$$

This is an equivalence relation because equality in  $K$  is reflexive, symmetric, and transitive. Exact-state return is the special case  $K = X$  and  $\kappa = \text{id}_X$ .

For each  $q \in Q$ , supply a nonempty family

$$\mathcal{P}_q \subseteq \text{End}(X) \times \text{End}(X).$$

A pair  $p = (f, r)$  describes a transformation  $f$  followed by a candidate return  $r$ . Its completed map is  $h_p = r \circ f$ . Define

$$H_q = \{r \circ f : (f, r) \in \mathcal{P}_q\}, \quad H = \bigcup_{q \in Q} H_q. \quad (2)$$

A successful return at  $x$  means

$$\kappa(r(f(x))) = \kappa(x), \quad \text{equivalently } h_p(x) \equiv_{\kappa} x. \quad (3)$$

The input may change:  $f(x) \neq x$  is compatible with a successful return. That inequality is required only when a particular test is intended to be a nontrivial perturbation. Neither every phase nor every admissible test must change every candidate.

#### Requirements that prevent a circular test

The state space, features, protocol maps, and application conditions must be specified before certifying success. A failed protocol remains a failed protocol; it is not removed by defining “admissible” to mean “successful.” A return map cannot be chosen after inspecting the answer merely to reset that particular candidate.

If recovery uses a stored copy, a memory, another participant, or a resource, that information and its cost belong in the state and protocol. Operations that fail to apply must be represented as failures or explicitly limited to a stated domain, not silently omitted from a universal claim. This edition uses total maps on an adequate state space for simplicity.

The identity features should also discriminate the alternatives relevant to the claim. A constant  $\kappa$  makes all states equivalent and every return pass. That is a mathematically valid but uninformative assessment.

## 4. Periodicity, fixed points, and recovery are different

The original definition of  $\text{Rec}(X)$  is repaired by renaming it:

$$\text{FinOrd}(X) = \{r \in \text{End}(X) : \exists k \in \mathbb{N}_{>0}, r^k = \text{id}_X\}. \quad (4)$$

The condition describes finite-order permutations, not recursive functions in computability theory. The exponent must be positive: allowing  $k = 0$  would make every endofunction qualify, since  $r^0 = \text{id}_X$ .

**Proposition 1.** *If  $r^k = \text{id}_X$  for some positive integer  $k$ , then  $r$  is bijective and  $r^{-1} = r^{k-1}$ .*

*Proof.* Both  $r \circ r^{k-1}$  and  $r^{k-1} \circ r$  equal  $r^k = \text{id}_X$ . Thus  $r^{k-1}$  is a two-sided inverse.  $\square$

The condition  $\exists k \forall x r^k(x) = x$  also imposes one common period. It is stronger, on an infinite set, than  $\forall x \exists k_x r^{k_x}(x) = x$ .

**Proposition 2.** *If  $|X| \geq 2$ , no point is fixed by every member of  $\text{FinOrd}(X)$ .*

*Proof.* Given  $x \in X$ , choose  $y \neq x$ . The transposition exchanging  $x$  and  $y$  and fixing every other point has order two and moves  $x$ . Therefore

$$\bigcap_{r \in \text{FinOrd}(X)} \text{Fix}(r) = \emptyset.$$

$\square$

This is why ALIGN must use a declared family of operations instead of every possible reversible map. It is also why  $r(x) = x$  cannot replace  $r(f(x)) = x$ : the former asks whether the return operation alone fixes a state, while the latter tests a completed transformation-and-return protocol.

On an infinite set, finite-order permutations need not even be closed under composition. On  $\mathbb{Z}$ , let  $a(n) = -n$  and  $b(n) = 1 - n$ . Then  $a^2 = b^2 = \text{id}$ , but  $(a \circ b)(n) = n - 1$  has infinite order. Closure under composition must therefore be constructed separately.

## I. INITIATE: make the contrast explicit

*The first distinction is the beginning of an identity question: what differs, and what is expected to remain recognizable through change?*

Define the contrast space by

$$\Delta = \{(a, b) \in U \times U : a \neq b\}. \quad (5)$$

This defines distinguishable pairs; it does not prove that a pair is minimal. A notion of minimality would require an additional order, complexity measure, or decomposition rule.

The requirement in v2 that some finite-order map sends  $a$  to  $b$  adds nothing if all such maps are allowed: the transposition of any unequal pair already does so. If only some transitions are physically or operationally available, their availability is extra model data.

For each perspective, choose a contrast-witness map  $c_q : X \rightarrow U \times U$ . Set

$$A_q(x) \iff c_q(x) \in \Delta. \quad (6)$$

If  $c_q(x) = (a_q(x), b_q(x))$ , this means  $a_q(x) \neq b_q(x)$ . Membership of both terms in  $U$  gives their common formal background. A causal or historical common origin would require a further relation; it is not established by set membership.

The INITIATE filter acts on a set  $S \subseteq X$ :

$$\nabla(S) = \{x \in S : \forall q \in Q, A_q(x)\}. \quad (7)$$

It selects candidates with recorded distinctions in every required perspective. Candidate generation, if needed, is a separate operation. The filter does not manufacture  $U$ , equality, or information from nothing.

### Four dimensions of initiation

For **Self**, identify the participant's relevant distinction. For **Relation**, identify the difference that co-ordination must span. For **Object**, identify observable alternatives. For **System**, identify distinct components, states, or arrangements.

These requirements are operational choices for an application of IDEA. A missing witness means that initiation is not established under that application. It does not prove that the candidate is meaningless, false, or nonexistent.

## II. DEFINE: state the structure and its identity

*Definition makes explicit what the contrast is a contrast of, and what a successful return must preserve.*

For each perspective, supply a structural predicate  $D_q : X \rightarrow \{\text{true}, \text{false}\}$ . It can check a stated grammar, a relation among components, a conservation constraint, or a consistency requirement. Its content must be given in the application; the word “coherent” alone is not a test.

Using the identity criterion from Equation (1), define

$$B = \{x \in X : \forall q \in Q, A_q(x) \wedge D_q(x)\}, \quad (8)$$

and the DEFINE filter

$$\mu(S) = S \cap B \quad (S \subseteq X). \quad (9)$$

Thus  $\mu(\nabla(S))$  retains initiated candidates satisfying the declared structural requirements. It does not assume that all future tests will succeed; that question belongs to ENGAGE and ALIGN.

The formula in v2 requiring  $x \neq y$  for *all*  $x, y \in \Delta$  is removed. Substituting  $y = x$  makes that requirement contradictory whenever  $\Delta$  is nonempty. Contrast belongs to the two members of a designated pair, not to every pair of variables regardless of whether they name the same object.

### Four dimensions of definition

**Self:** specify which internal organization or commitment is being tracked. **Relation:** state the shared expectations or compatibility conditions. **Object:** specify the relevant observable form. **System:** state the relations among components that constitute the organized whole.

Cogence, coherence, cohesion, and congruence remain useful names for questions about understanding, consistency, integration, and correspondence. They become mathematical conditions only when supplied with explicit predicates or measurements.

### Identity may be revised, but the claim then changes

The feature map  $\kappa$  may be learned or revised between assessments. Within one theorem or test report it is fixed. Changing it to hide a failed return creates a different assessment; it does not retroactively establish the earlier one. A model of learning may represent changing criteria in its state, but it must then specify the higher-level identity it is evaluating.

### III. ENGAGE: test declared transformations and returns

*To engage is to expose a proposed identity to a test whose outcome can matter.*

For every perspective, choose a nonempty test family  $\mathcal{E}_q \subseteq \mathcal{P}_q$ . Define

$$E_q(x) \iff \forall (f, r) \in \mathcal{E}_q, \kappa(r(f(x))) = \kappa(x). \quad (10)$$

Then

$$\varepsilon(S) = \{x \in S \cap B : \forall q \in Q, E_q(x)\}. \quad (11)$$

Each result concerns all tests in the stated engagement family, rather than merely the existence of one convenient pair of maps. Choosing only  $f = r = \text{id}_X$  would make a return test automatic and provide no evidence of robustness under change.

A test is *evaluated* only when its required predicate has been checked or proved. If its outcome is unavailable, record “unassessed”; do not count an empty record as a pass. The mathematical predicate has a definite interpretation in a fully specified model, while our evidence about it may be incomplete.

#### Four dimensions of engagement

**Self:** test whether the tracked internal organization remains recognizable after a challenge. **Relation:** test whether coordination or mutual expectations can be restored after a specified divergence. **Object:** test form or behavior under a declared intervention. **System:** test the organization of a composite after an alteration to its parts or connections.

Neither success nor failure carries an automatic metaphysical verdict. A failed return can expose a poor definition, an inadequate recovery procedure, missing information, a genuine boundary of stability, or an incorrect prediction. The observed failure is evidence about the stated model and claim.

#### The quantifiers must match the claim

The statement  $\forall f \exists r$  describes recoverability with an available choice of recovery;  $\exists r \forall f$  requires one recovery that works for every transformation. These are different assertions. This edition evaluates specified pairs  $(f, r)$  and universally quantifies over them. An adaptive recovery policy is permitted when its inputs, memory, and rule are explicitly encoded in the protocol.

## IV. ALIGN: establish invariance under a declared family

*Alignment is identity that survives every return required by the stated assessment.*

Let  $\mathcal{M} = \langle H \rangle$  be the monoid of all finite compositions of the completed maps in  $H$ , including the empty composition  $\text{id}_X$ . Define

$$I_* = \{x \in B : \forall h \in \mathcal{M}, \kappa(h(x)) = \kappa(x)\}. \quad (12)$$

This is the set of aligned candidates. It may be empty. Its members preserve the declared identity under every finite sequence of completed protocols, including sequences that mix perspectives.

For exact-state identity,  $\kappa = \text{id}_X$ , this becomes

$$I_* = B \cap \bigcap_{h \in \mathcal{M}} \text{Fix}(h), \quad \text{Fix}(h) = \{x \in X : h(x) = x\}. \quad (13)$$

Thus a fixed-point expression is available without demanding that a state resist every imaginable operation.

### What the condition does and does not preserve

Equation (12) checks identity at the ends of completed protocols. It does not require every intermediate state  $f(x)$  to be identical to  $x$ , or to meet every constraint in  $B$ . If intermediate safety, continued eligibility in  $B$ , resource preservation, or time limits matter, add them to the state, the tracked features, or explicit path conditions. Endpoint recovery alone does not prove those stronger properties.

For **Self**, alignment concerns the selected internal identity; for **Relation**, the selected relational pattern; for **Object**, the selected form or behavior; for **System**, the selected organization. In each case the result remains relative to the definition and protocol family.

### A finite test is not an unrestricted guarantee

Passing the engagement subset  $\mathcal{E}_q$  does not establish all of  $\mathcal{P}_q$ , unless those families coincide or a proof covers the remainder. Likewise, invariance under every finite composite does not assert convergence of an infinite trajectory. A limit claim needs additional topology, dynamics, and hypotheses.

## 5. A well-typed construction of return closure

Replace the phase hierarchy of v2 with a closure construction that keeps every element a function on the same state space:

$$\mathcal{M}_0 = \{\text{id}_X\} \cup H, \quad (14)$$

$$\mathcal{M}_{n+1} = \mathcal{M}_n \cup \{g \circ f : f, g \in \mathcal{M}_n\}, \quad n \in \mathbb{N}_0, \quad (15)$$

$$\mathcal{M} = \bigcup_{n \in \mathbb{N}_0} \mathcal{M}_n. \quad (16)$$

The sequence is increasing, and every  $\mathcal{M}_n$  is a subset of  $\text{End}(X)$ . The union includes every stage, including a terminal stage if one exists.

**Proposition 3.**  $\mathcal{M}$  is the smallest submonoid of  $\text{End}(X)$  containing  $H$ .

*Proof.* It contains  $\text{id}_X$  and  $H$ . If  $f \in \mathcal{M}_i$  and  $g \in \mathcal{M}_j$ , both are in  $\mathcal{M}_{\max(i,j)}$ , so  $g \circ f \in \mathcal{M}_{\max(i,j)+1}$ . Hence the union is closed under composition. Every submonoid containing  $H$  contains each stage by induction, and therefore contains the union.  $\square$

### Termination requires an assumption

If  $|X| = m < \infty$ , there are  $m^m$  endofunctions. Every strict increase adds at least one new map, so some finite  $N_*$  satisfies

$$\mathcal{M}_{N_*+1} = \mathcal{M}_{N_*}, \quad N_* \leq m^m - |\mathcal{M}_0|.$$

Once equality holds, subsequent stages are identical. This is a finite-state existence result, not an efficient algorithmic bound.

There need not be a finite stabilization index on an infinite space. On  $X = \mathbb{Z}$  with  $H = \{t\}$  and  $t(n) = n + 1$ ,

$$\mathcal{M}_n = \{t^k : 0 \leq k \leq 2^n\},$$

so every stage grows. Finite termination of v2's proposed phase and scope recursions was therefore not justified.

### Irreducibility must be relative to a composition rule

A generating set  $J$  for  $\mathcal{M}$  is *irredundant* when  $\langle J \rangle = \mathcal{M}$  and  $j \notin \langle J \setminus \{j\} \rangle$  for every  $j \in J$ . Testing whether a candidate lies outside  $\langle J \rangle$  is a precise novelty criterion. It considers arbitrary finite composites, not only two factors. It need not give a unique generating set, and implies no universal count of four phases or four scopes.

## 6. The corrected IDEA closure theorem

For feature-based identity, require each completed generator to respect the equivalence relation:

$$\forall h \in H, \forall x, y \in X, \quad x \equiv_{\kappa} y \implies h(x) \equiv_{\kappa} h(y). \quad (17)$$

This condition says that the same tracked identity cannot acquire different tracked identities merely because an untracked detail differs. It is automatic for exact equality, but must be checked or proved for a coarser feature map.

Define  $\text{Fix}_{\kappa}(h) = \{x \in X : h(x) \equiv_{\kappa} x\}$ .

**Theorem 1** (Closure under completed return protocols). *Assume Equation (17). For any  $x \in B$ ,*

$$x \in I_* \iff \forall h \in H, h(x) \equiv_{\kappa} x.$$

*Equivalently,*

$$I_* = B \cap \bigcap_{h \in H} \text{Fix}_{\kappa}(h). \quad (18)$$

*Proof.* The forward direction follows from  $H \subseteq \mathcal{M}$ . For the converse, use induction on the length of a word in the generators. The empty word fixes  $x$ . Suppose  $w(x) \equiv_{\kappa} x$ . For  $h \in H$ , Equation (17) gives  $h(w(x)) \equiv_{\kappa} h(x)$ , while the premise gives  $h(x) \equiv_{\kappa} x$ . Transitivity yields  $h(w(x)) \equiv_{\kappa} x$ . Every element of  $\mathcal{M}$  is a finite word, so the identity is preserved by all of  $\mathcal{M}$ .  $\square$

### Why the hypothesis matters

Let  $X = \{a, b, c\}$ , with  $\kappa(a) = \kappa(b) = 0$  and  $\kappa(c) = 1$ . Let  $h(a) = b$ ,  $h(b) = c$ , and  $h(c) = c$ . A one-step test at  $a$  passes:  $h(a) \equiv_{\kappa} a$ . But  $h^2(a) = c \not\equiv_{\kappa} a$ . The map does not respect the equivalence relation, since  $a \equiv_{\kappa} b$  but  $h(a) \not\equiv_{\kappa} h(b)$ .

Without Equation (17), retain the full definition in Equation (12); generator tests alone are insufficient. A numerical tolerance such as  $d(x, y) \leq \epsilon$  is generally not transitive and cannot simply replace the equivalence relation in this proof. Approximate recovery needs its own error bounds.

## 7. The corrected four-by-four grid

For  $q_j \in Q$ , define cumulative phase predicates:

$$C_{0j}(x) = A_{q_j}(x), \quad (19)$$

$$C_{1j}(x) = A_{q_j}(x) \wedge D_{q_j}(x), \quad (20)$$

$$C_{2j}(x) = C_{1j}(x) \wedge E_{q_j}(x), \quad (21)$$

$$C_{3j}(x) = C_{2j}(x) \wedge [\forall (f, r) \in \mathcal{P}_{q_j}, r(f(x)) \equiv_{\kappa} x]. \quad (22)$$

Then set

$$G_{ij} = \{x \in X : C_{ij}(x)\}, \quad i, j \in \{0, 1, 2, 3\}. \quad (23)$$

All sixteen cells are subsets of the same  $X$ , so their intersection is well defined. The cumulative predicates preserve the dependency of later phases on earlier phases. They are not claims of algebraic independence.

**Corollary 1** (Grid characterization). *Under the closure theorem's hypothesis,*

$$I_* = \bigcap_{i=0}^3 \bigcap_{j=0}^3 G_{ij}. \quad (24)$$

*Proof.* The intersection requires all initiation and definition predicates, giving membership in  $B$ , and all generator returns. Since  $\mathcal{E}_q \subseteq \mathcal{P}_q$ , the engagement predicates also hold. Equation (18) gives the result in both directions.  $\square$

### IDEA as a function, separately from its accepted set

Define filters  $\Phi_i : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$  by

$$\Phi_i(S) = S \cap \bigcap_{j=0}^3 G_{ij}.$$

Thus  $\Phi_0 = \nabla$ ,  $\Phi_1 = \mu$ ,  $\Phi_2 = \varepsilon$ , and  $\Phi_3 = \alpha$ . Define the assessment map

$$\text{IDEA}_{\mathfrak{M}} = \alpha \circ \varepsilon \circ \mu \circ \nabla. \quad (25)$$

Here  $\mathfrak{M}$  denotes the declared model data. Under the theorem's hypothesis,  $\text{IDEA}_{\mathfrak{M}}(S) = S \cap I_*$ . The map is monotone, idempotent, and *contractive*: its output is a subset of its input. It is a filter, not an extensive closure operator. By contrast,  $H \mapsto \langle H \rangle$  is an extensive closure operation on sets of transformations.

## 8. Reading and using the grid

The same formal predicate has a different application-level meaning in each perspective. The table below is an assessment guide, not sixteen automatic verdicts.

|                 | Self / I                                                 | Relation / We                                     | Object / It                                    | System / Its                                 |
|-----------------|----------------------------------------------------------|---------------------------------------------------|------------------------------------------------|----------------------------------------------|
| <b>Initiate</b> | Record an internal distinction.                          | Record a distinction between standpoints.         | Record observable alternatives.                | Record distinguishable arrangements.         |
| <b>Define</b>   | State the identity or commitment.                        | State expectations and compatibility.             | State the relevant form or behavior.           | State organizational constraints.            |
| <b>Engage</b>   | Evaluate a specified challenge and return.               | Evaluate a coordination test and return.          | Evaluate an intervention and return.           | Evaluate a composite-system test and return. |
| <b>Align</b>    | Establish identity preservation over declared protocols. | Establish preservation of the relational pattern. | Establish preservation of the chosen features. | Establish preservation of the organization.  |

### Record evidence rather than filling cells by analogy

For a particular  $x$ , a cell can be **established**, **failed**, or **unassessed**. These are evidential statuses, not three truth values in the classical mathematical model. An established cell needs its instantiated predicate and supporting computation, observation, or proof. A failed cell needs a witness. An unassessed cell remains open.

If a perspective has no meaningful interpretation in an application, record that limitation. Restricting to a subset of  $Q$  defines a reduced assessment, not a completed four-perspective assessment. The model does not turn a missing test into falsehood.

### Updating a model

Adding protocols may remove previously aligned candidates. If  $H \subseteq H'$  while  $X$ ,  $B$ , and  $\kappa$  are unchanged, then  $\langle H \rangle \subseteq \langle H' \rangle$ , hence

$$I_*(H') \subseteq I_*(H).$$

An earlier certificate remains valid for its earlier family, but it does not establish the enlarged claim. This gives the framework a direct way to accommodate new challenges without redefining failure as success.

## 9. Worked mathematical examples

### A. Exact recovery after nontrivial change

Let  $X = \{0, 1, 2\}$  and  $\kappa = \text{id}_X$ . In each perspective, use the protocol  $(f, r)$  where  $f$  exchanges 0 and 1 and fixes 2, and  $r = f$ . Choose  $U = \{u, v\}$ ,  $c_q(x) = (u, v)$ , and  $D_q(x) = \text{true}$  for all  $x, q$ , so  $B = X$ . Let  $\mathcal{E}_q = \mathcal{P}_q = \{(f, r)\}$ .

$$r \circ f = f^2 = \text{id}_X, \quad I_* = X.$$

The states 0 and 1 move during the transformation but return exactly. State 2 is unchanged. This demonstrates the mathematical distinction between change and successful return. Repeating one protocol in four perspectives is formally permitted but supplies no evidence that the perspectives are independent.

### B. A protocol that separates success from failure

Keep the same  $X$ ,  $\kappa$ , and  $B$ . In one perspective add the pair  $(d, \text{id}_X)$ , with  $d(0) = 0$ ,  $d(1) = 0$ , and  $d(2) = 2$ . Then

$$\text{Fix}(d) = \{0, 2\}, \quad I_* = \{0, 2\}.$$

State 1 fails this additional protocol. More strongly, no single map  $r : X \rightarrow X$  can exactly recover *both* 0 and 1 from  $d$ , because  $d(0) = d(1)$  would require  $r(0) = 0$  and  $r(0) = 1$  simultaneously. Recovery of the lost distinction requires extra information or a weaker identity criterion.

### C. Preserving a feature while the state changes

Let  $X = \{0, 1, 2, 3\}$ ,  $\kappa(x) = x \bmod 2$ , and  $h(x) = x + 2 \pmod{4}$ . The map  $h$  has no exact fixed points, but

$$\kappa(h(x)) = \kappa(x) \quad \text{for every } x \in X.$$

It respects the parity equivalence. Thus, with  $B = X$  and  $H = \{h\}$ , every state is aligned relative to parity and none is fixed under the nonidentity generator as an exact state. The result is specifically about parity; it does not show that every property of the state is preserved.

### What these examples establish

The repaired formalism admits nonempty results, meaningful failures, exact recovery, and feature preservation. Those examples establish the behavior of the model. They do not prove that invariance is identical to semantic truth.

## 10. Corrected example corpus: cases 1-4

The original sixteen cases are retained. Their unsupported checkmark grids are replaced by explicit logical assessments. Applying the formal grid would additionally require  $X$ ,  $\kappa$ , the predicates, and the protocol families for each case.

### 1. The liar paradox

**Claim:** “This statement is false.” With an unrestricted classical truth predicate, the intended self-reference leads to a truth value  $\ell$  satisfying

$$\ell = \neg\ell.$$

No Boolean value satisfies this equation. Calling the liar simply false is not a solution: on its intended reading, its falsity would make what it says true.

**Corrected status:** The unrestricted classical truth assignment is inconsistent; the sentence is not thereby proved “permanently false.” Notice that negation on  $\{0,1\}$  has period two. A cycle of truth-value changes can return, yet fail to solve  $\ell = \neg\ell$ . Recurrence alone does not resolve the paradox.

### 2. Gödel’s first incompleteness theorem

**Claim:** “This statement is unprovable in this system.” Fix an effectively axiomatized theory  $T$  extending sufficient arithmetic. A Gödel sentence is constructed to satisfy

$$T \vdash G_T \leftrightarrow \neg \text{Prov}_T(\ulcorner G_T \urcorner).$$

If  $T$  is sound for the standard natural numbers, then  $G_T$  is true there and unprovable in  $T$ . The Gödel-Rosser theorem obtains incompleteness from consistency alone, using a modified sentence. The theory, encoding, and hypotheses matter; a bare English self-reference is not the theorem. [1, Chapter 5]

**Corrected status:** A theorem about formal provability under stated assumptions. IDEA does not independently certify  $G_T$  or escape the theorem’s limitations.

### 3. “There is no truth”

If asserted as true and intended to include that very assertion, the unrestricted claim defeats itself under ordinary classical semantics. A restricted claim such as “none of these particular reports is true” need not do so.

**Corrected status:** The unrestricted self-inclusive assertion is self-undermining. Its failure does not prove IDEA’s particular phases, scopes, or universal necessity.

### 4. Identity tautology: “A is A”

For an object denoted by  $a$ ,  $a = a$  follows from the semantics of equality. For a proposition  $P$ ,  $P \rightarrow P$  is a propositional tautology. These are different grammatical uses of “A.”

**Corrected status:** Valid under the stated logical interpretation. The result does not require, and does not prove, a derivation of identity from recursive dynamics.

## 11. Corrected example corpus: cases 5-8

### 5. Contradiction: “A is not A”

With ordinary equality and a fixed referent,  $a \neq a$  is false in every interpretation. If the intended formula is instead  $P \wedge \neg P$ , it is unsatisfiable in classical propositional logic. Different times, respects, referents, or logics must not be silently identified.

**Corrected status:** Classically false or unsatisfiable under the precise reading above. Failure is established by the logical semantics, not by a missing return path.

### 6. Self-reproducing function

“This function outputs itself” must distinguish a program from its encoded description. For a program-text string  $p$ , a quine satisfies

$$\text{Run}(p) = p,$$

where Run means execution under a specified interpreter and the output is compared as text. A function returning a reference to itself is a different construction.

**Corrected status:** A particular self-reproduction claim can be established by a construction or execution trace. It does not make the program, all its outputs, or all claims about it universally true. The original example supplied no program to verify.

### 7. “Aliens exist”

This is an empirical existence claim whose meaning depends on such matters as what counts as life and the intended range of places and times. The absence of a completed IDEA assessment supplies neither a proof nor a disproof.

**Corrected status:** Unresolved by the formalism alone. Evidence is required; lack of a return certificate does not establish falsehood or inherent untestability.

### 8. “Identity emerges from recursion”

Within the corrected model, a *stable identity relative to  $\kappa$  and  $\mathcal{M}$*  is defined by Equation (12). An application can investigate how patterns meeting that definition form or persist.

That does not derive the equality relation used to define sets and maps. Nor does it establish that all physical, personal, or semantic identity has the same origin. A causal account of the emergence of a pattern needs dynamics and evidence beyond a definition of its invariance.

**Corrected status:** A legitimate modeling definition and a broader research hypothesis, depending on the reading. The universal ontological claim remains unproved.

## 12. Corrected example corpus: cases 9-12

### 9. There is no largest prime

Assume that  $p_1, \dots, p_n$  are all the primes. Set

$$N = p_1 p_2 \cdots p_n + 1.$$

Since  $N > 1$ , it has a prime divisor  $p$ . For every  $i$ , division of  $N$  by  $p_i$  leaves remainder 1, so  $p$  is absent from the list. This is a contradiction. The number  $N$  itself need not be prime.

**Corrected status:** A proved mathematical theorem. Its proof supplies the justification; a four-by-four pattern of checkmarks would not replace that proof.

### 10. Law of non-contradiction

In classical propositional logic,

$$\neg(P \wedge \neg P)$$

is true for both Boolean values of  $P$ . “Nothing can both be and not be” needs that same proposition, interpretation, time, and respect to be meaningful as this logical law.

**Corrected status:** Classically valid. Other logical systems require their own semantics; a claim about all such systems does not follow from the classical truth table.

### 11. Digital simulation hypothesis

“We live in a simulation” leaves the simulator, implementation, and empirical consequences unspecified. Particular simulation proposals might have observable consequences; direct access to a simulator is not logically necessary for every possible test.

**Corrected status:** The broad claim is underdetermined here. It is not shown false by a supposed failure of “systemic return,” and no formal IDEA certificate has been supplied.

### 12. Axiom of Choice

For every *set-indexed* family  $(A_i)_{i \in J}$  of nonempty sets, the Axiom of Choice asserts that there is a function  $c$  with domain  $J$  such that

$$c(i) \in A_i \quad \text{for every } i \in J.$$

It is an axiom of ZFC and is independent of ZF if ZF is consistent. Thus it holds in every model of ZFC; IDEA has not proved it from weaker assumptions. [2] [3]

**Corrected status:** Assumed in ZFC; independent of ZF under the consistency condition. “Recursively coherent” is not a substitute for specifying the background axioms.

## 13. Corrected example corpus: cases 13-16

### 13. Circular definition: “X is defined by X”

The equation  $x = x$  is satisfied by every  $x$  in its domain. If the domain has more than one element, it does not determine a unique one. It is uninformative as a standalone definition, rather than false. Recursive equations can define objects when their additional constraints establish existence and, where needed, uniqueness.

**Corrected status:** An insufficient definition as stated. Circular syntax by itself proves neither contradiction nor a successful construction.

### 14. Schrödinger’s cat and quantum measurement

An idealized entangled state has the form

$$|\Psi\rangle = \alpha|\text{alive}\rangle|E_a\rangle + \beta|\text{dead}\rangle|E_d\rangle, \quad |\alpha|^2 + |\beta|^2 = 1.$$

Decoherence suppresses interference in the reduced description as the environmental states become nearly orthogonal. That does not, by itself, select a unique outcome or prove that the original state has been recovered. Outcome claims require the relevant measurement framework or interpretation. [4, Sections II and IV]

**Corrected status:** The original claim that this physically demonstrates IDEA’s permanent recursive closure is unsupported. A superposition is also not the classical proposition  $P \wedge \neg P$ .

### 15. Modal collapse: “Everything that could happen must happen”

Read this as  $\Diamond P \rightarrow \Box P$ . Consider two mutually accessible worlds  $u, v$ , with  $P$  false at  $u$  and true at  $v$ . At  $u$ ,  $\Diamond P$  is true and  $\Box P$  is false. The implication therefore fails in this model, even on an S5 frame.

**Corrected status:** Not valid in standard normal modal logics admitting that frame. It is not a contradiction in every model: a one-world reflexive frame validates the schema. Modal possibility and necessity must not be conflated.

### 16. Ontological grounding: “Existence exists”

The phrase needs an interpretation. If formalized as  $\exists x(x = x)$ , it is valid in ordinary first-order semantics with nonempty domains. If empty domains are permitted, it is false there.

**Corrected status:** A consequence of the chosen nonempty-domain semantics under that translation. It does not independently prove a particular primordial field or a universal theory of existence.

## 14. Permanence, truth, and the limits of the result

IDEA can organize a disciplined investigation of a candidate’s identity through transformation. Its mathematical result is a conditional invariance claim. Four other questions must remain distinct:

| Question                            | Required justification                                                  |
|-------------------------------------|-------------------------------------------------------------------------|
| Is a proposition true in a model?   | An interpretation and its satisfaction relation.                        |
| Is it provable from a theory?       | A derivation using the theory’s axioms and rules.                       |
| Does a state preserve its identity? | A specified identity criterion, operations, and an invariance argument. |
| Is an empirical claim supported?    | Observations or experiments linked to the claim by an adequate model.   |

A representation of a false statement can remain perfectly stable under repeated copying. A true report can change its physical encoding. Invariance of a representation is therefore neither sufficient nor necessary for the truth of its content without additional assumptions connecting those notions.

If an application represents a formula  $\varphi$  inside  $x$ , the condition  $x \in I_*$  does not by itself imply  $\mathcal{A} \models \varphi$  for an interpretation  $\mathcal{A}$ . To obtain that implication, one must prove a soundness connection between the assessment predicates and the intended semantics. Defining a predicate to mean “is true” supplies an assumption or an oracle, not an independent method for deciding truth.

### The universal claims retained as hypotheses

The philosophical proposal that contrast, definition, testing, and return are necessary for every intelligible inquiry can be argued for and investigated. It is not established by the closure theorem. A critic’s use of language or distinctions does not entail that the critic accepts IDEA’s specific architecture, and a failed candidate does not automatically confirm the framework.

The former “unfalsifiability corollary” is therefore withdrawn as a mathematical result. Definitions are not empirical predictions, but models and applications built with them can still be challenged by incorrect predictions, unsuitable identity criteria, missing scopes, or failed returns. A return-based explanation must make contact with those possible failures to earn evidential support.

*To initiate is to make a distinction. To define is to state what is being preserved. To engage is to expose that claim to a test. To align is to establish the return that the stated conditions actually support.*

## Appendix A. Correction record against v2

Page numbers in this appendix refer to the supplied 28-page v2. The revised text is a complete replacement of its mathematical exposition, with the conceptual architecture and all sixteen examples retained.

| v2         | Problem                                                                                                          | Repair                                                                                                                                 |
|------------|------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| 1-2, 7     | Universal necessity and unfalsifiability presented as proved.                                                    | Retained only as philosophical hypotheses. The mathematical result is relative invariance, not universal semantic truth.               |
| 2          | $\text{Rec}(X)$ misnames finite-order permutations; $k = 0$ is not excluded.                                     | Renamed $\text{FinOrd}(X)$ with $k > 0$ . General dynamics use declared endofunctions.                                                 |
| 2          | “Minimal” contrasts are not shown minimal; the return condition is automatic using transpositions.               | Define $\Delta$ as unequal ordered pairs. Minimality and restricted reachability require extra structure.                              |
| 2-3        | $\Psi_n$ begins as contrasts and becomes functions on functions, then is treated as a single state space.        | All states stay in $X$ and all maps in $\text{End}(X)$ . Composition closure is defined by the increasing sequence $\mathcal{M}_n$ .   |
| 3          | Four phases claimed to emerge without a derivation or an irreducibility proof.                                   | Four named phase filters are explicit modeling choices. No theorem of uniqueness or minimality is claimed.                             |
| 3-5        | Scope equivalence is unspecified; equivalence classes are composed as functions; return motifs overlap.          | Four labels are supplied by the chosen axes. Protocol families may overlap. No unproved partition is used.                             |
| 3          | The Return Validity Criterion compares a map with only two factors and mixes maps with scope classes.            | Relative novelty is membership outside a generated monoid; irredundancy is stated separately for actual maps.                          |
| 4          | Finite stabilization is assumed and $\text{Rec}(\Psi)$ is redefined as stable phase outcomes.                    | Use one consistent meaning per symbol. Prove finite stabilization only for finite $X$ ; provide an infinite counterexample.            |
| 5-6        | $I_{\text{final}}$ means both an index and a set.                                                                | Separate $N_*$ , when finite stabilization exists, from the aligned set $I_*$ .                                                        |
| 5-6, 26-27 | Every point must be fixed by every allowed finite-order map.                                                     | Show that the unrestricted condition is empty for $ X  \geq 2$ . Quantify over specified completed protocols instead.                  |
| 5-6        | Global invariance and invariance in all parts of its partition are redundantly presented as distinct conditions. | The grid is explicitly cumulative. A generator theorem supplies the actual compositional result.                                       |
| 6          | A lambda expression intersects arbitrary state elements, ignores its input, and conflates a function with a set. | $\text{IDEA}_{\mathcal{M}} : \mathcal{P}(X) \rightarrow \mathcal{P}(X)$ filters candidate sets; $I_* \subseteq X$ is its accepted set. |

| v2    | Problem                                                                                                                          | Repair                                                                                                                                      |
|-------|----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 6     | “Resolves under a phase” has no defined predicate.                                                                               | Supply $C_{ij}$ and $G_{ij}$ with fixed types, quantifiers, and dependencies.                                                               |
| 7-15  | Checkmarks supply verdicts without instantiated models or proofs.                                                                | Retain all sixteen examples with corrected logical status and explicit reasons; do not fabricate grid results.                              |
| 17-18 | $\nabla$ is both $U \rightarrow \{a, b\}$ and a set constructor; $a \sim U \sim b$ and $\text{Rec}(q)$ are not adequately typed. | $\nabla$ is a filter on $\mathcal{P}(X)$ . Contrast witnesses lie in $U^2$ ; background membership is distinguished from causal origin.     |
| 20-21 | $\forall x, y \in \Delta, x \neq y$ contradicts the case $x = y$ ; pair and state notation changes.                              | Unequal components occur only within specified contrast witnesses. Structural predicates operate on $x \in X$ .                             |
| 23-24 | Existence of one pair $f, r$ is treated as success under all transformations; $T$ and domains are unclear.                       | Declare protocol families before evaluating them and universally quantify over those families. All maps act on $X$ .                        |
| 24-27 | ENGAGE and ALIGN share incompatible uses of $\iota$ ; mathematical and ontological conclusions are conflated.                    | Use distinct phase filters $\varepsilon$ and $\alpha$ . Make evidence, return, and semantic truth separate.                                 |
| 28    | The theorem requires $\psi_i(x) \neq x$ and $r(x) \neq x$ , the opposite of fixed-point closure.                                 | Require $r(f(x)) \equiv_{\kappa} x$ for declared protocols; prove extension to composites. No universal phase-change inequality is imposed. |
| 28    | “No new phase can be added without contradiction” is treated as finite algorithmic convergence.                                  | Closure is equality of successive stages in a specified recurrence, when reached. It is not a proof that no broader model is possible.      |

## Appendix B. Additional checks and remaining obligations

### A direct counterexample to the old phase derivation

Take  $U = \{0, 1\}$ . The old contrast set has two elements,  $(0, 1)$  and  $(1, 0)$ . Its proposed first phase set consists only of the transposition of those two contrasts. The next stage is empty, because a singleton has no nonidentity permutation. Subsequent stages remain empty. The cardinalities are

$$2, 1, 0, 0, \dots$$

That construction neither accumulates four phases nor proves that four mutually irreducible transformations exist. Its union also mixes contrast objects with functions on those objects. Merely repairing a typographical sign would leave these defects intact.

### Finite checks used in this revision

The small-state examples, the empty universal-fixed-point counterexample, the distinction between exact and feature-based return, and the necessity of the equivalence-respecting hypothesis were checked computationally. The generator theorem was also checked against all 378 unordered pairs (allowing repetition) of the 27 maps on a three-element state space and all five equivalence relations. All 1,116 cases satisfying its hypothesis passed; 414 cases without that hypothesis showed why the restriction can matter. These checks supplement the proofs; they do not replace them or establish an empirical claim about the world.

### What remains to be supplied in an actual application

1. Define  $U$ ,  $X$ , and any encoding of types, context, memory, and resources.
2. Specify the contrast witnesses  $c_q$ , structural predicates  $D_q$ , and identity features  $\kappa$ .
3. Declare the transformation-and-return pairs in each  $\mathcal{P}_q$ , together with an engagement subset  $\mathcal{E}_q$  and its evidence.
4. Prove Equation (17) if generator checks are to stand for every composite; otherwise assess the full monoid criterion.
5. State whether the result concerns exact recovery, feature preservation, an approximate metric, or a stronger condition on entire paths.
6. Provide any further argument connecting the return criterion to an empirical prediction or a semantic truth claim.

The corrected theorem creates a consistent formal target for these choices. It does not select them uniquely. The proposed connection between IDEA and the necessary structure of all intelligibility remains additional philosophical work.

## References and source basis

**User-supplied sources.** *Canon of Permanent Truth (IDEA)* (5 pages) and *Canon of Permanent Truth (IDEA) v2* (28 pages), attributed in the documents to Discord @unicorn.lo. The original cover artwork is reproduced from v2. The first version supplies the conceptual comparison; v2 is the revision target.

[1] Open Logic Project. *Incompleteness and Computability: An Open Introduction to Gödel's Theorems*. Chapter 5, especially the first incompleteness theorem, Rosser's theorem, and the comparison with Gödel's original assumptions. [Open textbook PDF](#).

[2] Kurt Gödel. "The Consistency of the Axiom of Choice and of the Generalized Continuum-Hypothesis." *Proceedings of the National Academy of Sciences* 24 (1938), 556-557. [doi:10.1073/pnas.24.12.556](#); [open archive](#).

[3] Paul J. Cohen. "The Independence of the Continuum Hypothesis, II." *Proceedings of the National Academy of Sciences* 51 (1964), 105-110. [doi:10.1073/pnas.51.1.105](#). This is part of the forcing work establishing the relevant independence results.

[4] Maximilian Schlosshauer. "Decoherence, the measurement problem, and interpretations of quantum mechanics." *Reviews of Modern Physics* 76, 1267-1305, published 2005. See especially Sections II and IV. [Author's preprint](#); [doi:10.1103/RevModPhys.76.1267](#).

The elementary set, function, modal-countermodel, and prime-number arguments used in this revision are shown directly in the text. The external references support the specialized examples; they do not independently endorse IDEA or the proposed formal repair.

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Edition note: v2.1 replaces the mathematical construction and revises conclusions that depended on it. It preserves the four phases, four perspectives, conceptual emphasis on identity through change, and all sixteen example topics. It does not claim that the new assumptions were derived in v2.